

SIMEON NEDELICHEV

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PROFESSIONAL EXPERIENCE

Researcher & Senior RL Engineer

2026 – Present

Institute of Artificial Intelligence

Developing learning-based whole-body control, data-driven motion generation, and estimation methods for humanoid and articulated robots. Building simulation pipelines, motion libraries, and deployable control stacks with academic and industry partners.

Senior Optimal Control & RL Engineer

2023 – 2026

Sber Robotics Center, Sber

Developed the RL whole-body control (WBC) and locomotion policies demonstrated on the Green humanoid at AIJ Conference 2025. Focused on control, with modeling, MuJoCo simulation, and training/evaluation pipelines; also performed motor and actuator system identification.([video](#)).

Senior Lecturer

2022 – Present

Institute for Robotics and Computer Vision, Innopolis University

Teaching advanced courses in robotics, mathematical modeling, motion planning, and applied and nonlinear control. Curriculum development and student mentoring in dynamics, simulation, and embodied AI.

Junior Researcher

2019 – 2023

Center for Technologies in Robotics and Mechatronics Components, Innopolis University

Mathematical modeling, identification, and control of twisted-string actuators; optimal nonlinear control and trajectory optimization; mechanical and electrical prototyping of novel robotic systems.

Research Assistant

2017 – 2019

BioRobotics Laboratory, Korea University of Technology and Education

Analysis, prototyping, and nonlinear control of twisted-string actuators (TSA); hardware implementation of advanced control algorithms.

EDUCATION

Ph.D. studies in Computer Science (Robotics track), completed

2019 – 2023

Innopolis University

GPA 4.87/5 (97/100). Institute of Robotics and Computer Vision.

Thesis: *Dynamic parameter estimation and adaptive control over robotic systems: generalized momentum approach.*

M.Sc. in Mechanical Engineering

2019

Korea University of Technology and Education

GPA 4.45/4.5 (98/100).

Thesis: *Design of Robotic Gripper with Constant Transmission Ratio Based on Twisted String Actuator.*

M.Sc. in Robotics & Mechatronics

2018

Moscow State Technological University STANKIN

GPA 4.9/5 (98/100).

Thesis: *The Dynamics of Controlled Motion of Industrial Robots.*

B.Sc. in Robotics

2016

Moscow State Technological University STANKIN

GPA 4.35/5 (87/100).

Thesis: *Development and research of a dynamic model of a robot manipulator.*

SELECTED RESEARCH & SOFTWARE PROJECTS

WBC-Gen: Reactive Generative Motion Planner

Fast proprio-conditioned latent flow-matching planner that emits trackable WBC references for a frozen whole-body controller; generalizable across tasks under one unified WBC contract; can [recover and stand up from failed states](#) without a fall/stand mode FSM. Gen and WBC run together on the Unitree G1 onboard CPU, with [walk/sprint teleop](#), [disturbance rejection](#), and [agile recovery](#) on hardware. [Try Gen alongside WBC tracking](#) (WASD to steer, Shift to sprint; G switches clips $\leftrightarrow$ Gen); deploy code in [wbc-g1-deploy](#); training code and paper coming soon.

WBC-Mjlab: Whole-Body Control in MuJoCo Lab ([project page](#)) — [web demo](#)

Shared MDP for whole-body motion tracking in mjlab: multi-clip training, paper-specific task presets (ZEST, BeyondMimic-style RSI, and related setups), motion-data pipeline, and sim-to-real export for humanoids.

PSM: Predictive Style Matching for Natural Locomotion ([project page](#))

Offline predictor maps lower-body state and velocity commands to upper-body and gait targets during RL training; deployed on the Unitree G1 with task-only inference at runtime. arXiv:2606.07083 (2026).

Locomotion and Whole-Body Control for Sber Green ([AIJ 2025 presentation](#))

Developed the RL locomotion and whole-body control policies demonstrated on the Green humanoid at AIJ 2025, improving gait naturalness; RL whole-body control trained on synthetically generated reference trajectories (without mocap capture or retargeting) with unified tracking for teleoperation, dance, and zero-shot skills.

OPEN SOURCE CONTRIBUTIONS

Pinocchio ([stack-of-tasks/pinocchio](#)) — contributor to the rigid-body dynamics library (inertia parametrizations and physical consistency).

Pink ([stephane-caron/pink](#)) — contributor to differential inverse kinematics for Pinocchio (tasks, constraints, and examples).

Mink ([kevinzakka/mink](#)) — contributed `EqualityConstraintTask` for closed-chain mechanisms; humanoid and manipulation examples in the MuJoCo port of Pink.

GMR: General Motion Retargeting ([YanjieZe/GMR](#)) — contributor (SOMA BVH format and multi-robot configs).

mujoco-sysid ([based-robotics/mujoco-sysid](#)) — contributor; system identification in MuJoCo and MJX (parameter estimation, dynamics regressors).

MJINX: Differentiable GPU-accelerated Numerical IK ([based-robotics/mjinx](#)) — co-author (JAX and MuJoCo MJX).

PUBLICATIONS

Preprints

- **Nedelchev S**, Chaikovskaia E, Davydenko E, Zaliaev E, Gorbachev R. Predictive Style Matching: Natural and Robust Humanoid Locomotion. arXiv:2606.07083, 2026.
- Maslennikov E, Zaliaev E, Dudorov N, Shamanin O, Karanov D, Afanasev G, Burkov A, Lygin E, **Nedelchev S**, Ponomarev E. Robust RL Control for Bipedal Locomotion with Closed Kinematic Chains. arXiv:2507.10164, 2025.
- Alentev I, Kozlov L, Domrachev I, **Nedelchev S**, Ryu JH. VIMPPI: Enhancing Model Predictive Path Integral Control with Variational Integration for Underactuated Systems. arXiv:2505.05507, 2025.

Journals

- **Nedelchev S**, Kozlov L, Khusainov R, Gaponov I. Enhanced Adaptive Control over Robotic Systems via Generalized Momentum Dynamic Extensions. *Russian Journal of Nonlinear Dynamics*, 19(4):633–646, 2023.
- Fam CA, **Nedelchev S**. Optimization Driven Robust Control of Mechanical Systems with Parametric

Uncertainties. *Russian Journal of Nonlinear Dynamics*, 19(4):585–597, 2023.

- Skvortsova V, **Nedelchev S**, Brown J, Farkhatdinov I, Gaponov I. Design, characterisation and validation of a haptic interface based on twisted string actuation. *Frontiers in Robotics and AI*, 2022.
- **Nedelchev S**, Skvortsova V, Guryev B, Gaponov I, Ryu JH. On Energy-Preserving Motion in Twisted String Actuators. *IEEE Robotics and Automation Letters*, 2021.
- **Nedelchev S**, Gaponov I, Ryu JH. Accurate dynamic modeling of twisted string actuators accounting for string compliance and friction. *IEEE Robotics and Automation Letters*, 2020.

Conference papers

- Jnadi A, Khusainov R, **Nedelchev S**, Savin S. Explicit Model Predictive Control Design based on Constrained Zonotope Propagation. *DCNA*, 2023.
- **Nedelchev S**, Kirsanov D, Gaponov I, Seong H, Ryu JH. On Smooth Time-Optimal Trajectory Planning in Twisted String Actuators. *ICRA*, 2021.
- Sabirova A, **Nedelchev S**, Gaponov I. Parameter Identification in Mechanical Systems with Energy-Based Regressor: Preliminary Study. *NIR*, 2021.
- **Nedelchev S**, Kirsanov D, Gaponov I. IMU-based Parameter Identification and Position Estimation in Twisted String Actuators. *IROS*, 2020.
- Balakhnov O, **Nedelchev S**, Gaponov I. Preliminary Study on Slack-Free Model Predictive Control of Twisted String-Based Antagonistic Joints. *NIR*, 2020.
- **Nedelchev S**, Gaponov I, Ryu JH. High-bandwidth control of twisted string actuators. *ICRA*, 2019.
- **Nedelchev S**, Gaponov I, Ryu JH. Design of robotic gripper with constant transmission ratio based on twisted string actuator. *IROS*, 2018.
- Kosterev D, Vorotnikov A, **Nedelchev S**, Romash E, Poduraev Y. Development of 2-DoF Adaptive Mechatronic Device with Corrective Adjustment of Laser Tracker Reflector for Industrial Robot Calibration. *Annals of DAAAM & Proceedings*, 28, 2017.

TEACHING

Innopolis University — Senior Lecturer (2022–present), Teaching Assistant (2019–2022)

FORC: Fundamentals of Robot Control ([course](#); [cite](#); [repository](#))

Introductory course on state-space modeling, stability, linear and nonlinear control, and feedback linearization for robotic systems (lectures and Colab/Python labs).

Modeling and Simulation of Robotic Systems ([course](#); [cite](#); [repository](#))

Modeling, simulation, and analysis of articulated robots (lectures and Python labs).

- Applied Nonlinear Control; Fundamentals of Robot Control
- Modeling and Simulation of Robotic Systems; Linear Control Theory; Robotic Systems
- Computational Intelligence; Advanced Robotics

TECHNICAL SKILLS

Programming: Python, C/C++, Bash

Robotics & ML: MuJoCo, mlab, Isaac Sim, Isaac Lab, RL/WBC training pipelines, system identification

Embedded: STM32, ARM, ESP32, RP2040 (MicroPython, HAL, FreeRTOS, mbed)

Tools: Linux, LaTeX, Docker, uv, pixi

HONORS

- Best Master Thesis Award, Korea University of Technology and Education, 2019
- First Prize, Russian thesis competition “Be-First” (Computer & Information Technologies), 2018

RESEARCH INTERESTS

Articulated robots; reinforcement learning; hybrid methods; sim-to-real transfer; motion imitation; whole-body control and humanoid locomotion; learning-based and nonlinear control; system identification and estimation; differentiable simulation; data-driven motion generation.